
An Agentic LLM Copilot for Aircraft Turnaround Monitoring and Delay Recovery

Student Name: Seif Eldin Tourky

Outline

- ▶ Context — aircraft turnaround and A-CDM
- ▶ Problem statement — why coordination is hard
- ▶ Objectives — what the system will do
- ▶ Methodology — simulator, agent, baselines
- ▶ Project management — plan and progress
- ▶ Literature review — three converging fields
- ▶ Next steps — what remains and key risks

Context

- ▶ Aircraft turnaround = ground phase between arrival on stand and departure
 - ▶ Low-cost short-haul turnarounds completed in 25–35 minutes
 - ▶ Parallel + sequenced tasks: deboarding, cleaning, fuelling, catering, loading, boarding, pushback
- ▶ Coordinated under A-CDM (Airport Collaborative Decision Making)
 - ▶ EUROCONTROL / ACI EUROPE / IATA / CANSO framework
 - ▶ Milestones Approach — 16 milestones; partially, not strictly, sequential
 - ▶ Key target time: TOBT → drives TSAT and TTOT
- ▶ The turnaround coordinator continuously fuses events, predicts breaches, and decides recovery actions — **an agentic decision problem**

Problem Statement — WHY

- ▶ Coordinators make rapid, high-stakes decisions with **no decision support beyond status dashboards**
 - ▶ A dashboard reports that a milestone is **late** — it does not reason about what that lateness will **cause**
 - ▶ Propagation reasoning is done **mentally, in real time**, often across several aircraft
- ▶ **Who is affected**
 - ▶ Ground handlers (OTP targets, contractual exposure); airlines (delay cost); airports/ANSPs (slot and stand predictability); passengers
- ▶ **Where / when**
 - ▶ On stand, during every turnaround — most acute in high-frequency low-cost operations
- ▶ **Why it matters:** reactionary delay propagates along aircraft rotations — one poorly recovered turnaround degrades punctuality network-wide

Objective — WHAT

- ▶ **Main objective:** design, build and evaluate an agentic LLM copilot that ingests a live turnaround milestone event stream, predicts off-block breaches, reasons about delay propagation, and recommends explainable recovery actions
- ▶ **Sub-objectives**
 - ▶ Build a configurable A-CDM-grounded turnaround simulator with controlled disruption injection
 - ▶ Implement rule-based and critical-path baselines for comparison
 - ▶ Design an agent with a perceive–predict–plan–act–reflect loop and tool use
 - ▶ Generate ranked, explainable recovery recommendations
 - ▶ Evaluate against baselines and characterise failure modes
- ▶ **Positioning:** decision support — the agent recommends, the human decides (safety-critical domain)

Methodology — HOW

▶ **Simulation-based experimental study**

- ▶ Turnaround modelled as a dependency graph: tasks, durations, precedence, resource limits
- ▶ Simulator generates milestone event streams; disruptions seeded and repeatable

▶ **Agent architecture — LLM as orchestrator**

- ▶ ReAct-style loop; tools: milestone query, propagation estimator, critical-path solver, comms drafter

▶ **Evaluation**

- ▶ Quantitative: on-time performance, milestone adherence, prediction lead time and error vs baselines
- ▶ Qualitative: expert rating of recommendation quality and explanation usefulness

▶ **Why these methods**

- ▶ Real operational data is **commercially restricted** — simulation gives ground truth, control and repeatability
- ▶ Classical solver handles exact scheduling; LLM handles ambiguity, explanation and communication

Project Management

- ▶ Plan: 8 work packages across the project period, tracked as weekly sprints
 - ▶ WP1 Literature review & scope • WP2 Simulator • WP3 Baselines
 - ▶ WP4 Agent + tool loop • WP5 Recovery recommendations
 - ▶ WP6 Evaluation • WP7 Expert assessment • WP8 Analysis & write-up
- ▶ **Risk management built into the plan**
 - ▶ Simulator fidelity check = go/no-go gate before agent build
 - ▶ Extensions (comms drafting, expert study) marked **cuttable** if core work slips
 - ▶ Final week reserved as buffer for write-up and submission

Gantt Chart

MSc AI Project — Gantt Chart

Agentic LLM Copilot for Aircraft Turnaround Monitoring & Delay Recovery

Period: 10 Jun – 1 Aug 2026 • ~12.5 h/week (part-time) • Deadline: 1 Aug 2026

#	Task	Type	Start	End	Wks	Est. hrs	9 Jun	16 Jun	23 Jun	30 Jun	7 Jul	14 Jul	21 Jul	28 Jul
1	Literature review & finalise scope	Core	10 Jun	20 Jun	1.4	18								
2	Build turnaround simulator (A-CDM milestones, dependency graph, disruption injection)	Core	16 Jun	29 Jun	1.9	23								
3	Implement baselines (rule-based + critical-path)	Core	23 Jun	4 Jul	1.6	20								
M	★ Simulator fidelity check (validate against domain knowledge)	Milestone	29 Jun	29 Jun	0	13								
4	Agent architecture: perceive–predict–plan–act loop + tool use	Core	30 Jun	13 Jul	1.9	23								
5a	Recovery recommendation logic	Core	7 Jul	16 Jul	1.3	16								
5b	Stakeholder comms drafting (extension)	Extension	16 Jul	22 Jul	0.9	13								
6	Evaluation experiments (N seeded disrupted turnarounds vs baselines)	Core	13 Jul	23 Jul	1.4	18								
7	Expert assessment of recommendations (extension)	Extension	21 Jul	27 Jul	0.9	13								
8	Analysis & dissertation write-up	Buffer	16 Jul	31 Jul	2.1	27								

Buffer week (28 Jul–1 Aug): proofread, final fixes, submission. Extensions (5b, 7) are cuttable if core slips.

Hours budget check

Total estimated hours (core + extensions):

184

Available hours (7.5 weeks × 12.5 h):

94

Legend

	Core (must finish)
	Cuttable extension
	Buffer / write-up
	Milestone

Literature Review (1/2)

▶ Three converging fields

▶ 1. Agentic LLMs — planning, tool use, reflection

- ▶ ReAct (Yao et al.) — interleaving reasoning and acting; foundation of the agent loop
- ▶ Toolformer (Schick et al.); tool-learning surveys — models invoking external APIs
- ▶ Planning surveys (Huang et al.) — decomposition, plan selection, external modules, reflection
- ▶ LLM + classical planner work (PDDL) — precedent for orchestrating a solver
- ▶ **Known weakness:** brittleness — error propagation, unreliable tool selection, poor recovery from own mistakes

▶ 2. Aircraft turnaround & A-CDM

- ▶ EUROCONTROL A-CDM specification — milestones, TOBT/TSAT logic, alert thresholds
- ▶ Provides a documented, industry-standard task model — not an invented one

Literature Review (2/2)

▶ 3. Delay prediction and propagation

- ▶ Dominated by classical ML: gradient boosting (XGBoost/LightGBM), random forests, quantile regression
- ▶ Turnaround milestone prediction — cascaded gradient boosting for A-CDM
- ▶ Propagation modelling — chained prediction along aircraft rotations; primary vs reactionary delay
- ▶ Recent work embeds ML-derived uncertainty into prescriptive optimisation for proactive buffer planning
- ▶ Computer-vision turnaround monitoring detects milestone timestamps from apron video

▶ The gap

- ▶ Classical work predicts delay **magnitude** well, but offers little **explainable, action-level recovery recommendation**
- ▶ Agentic LLMs are **essentially absent** from live turnaround coordination
- ▶ This project bridges the two fields — and asks what the agentic layer adds over classical scheduling

Next Steps

- ▶ **Progress to date** — Completed: literature review, scope finalised, simulator design
 - ▶ In progress: simulator implementation
- ▶ **Remaining work**
 - ▶ Complete simulator and validate fidelity against domain knowledge
 - ▶ Implement baselines, then agent loop with tool use
 - ▶ Run evaluation experiments; analyse and write up
- ▶ **Risks and concerns**
 - ▶ **Scope: simulator must stay minimal — fidelity vs time trade-off**
 - ▶ **Agent brittleness may limit recovery quality — mitigated by decision-support framing**